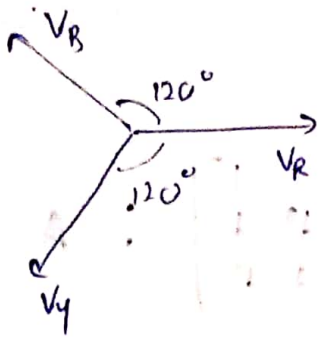
2019

Q1)

$V_R = 0.1\angle 0^\circ$, $V_B = 1.1\angle 150^\circ$, $V_Y = 1.1\angle -150^\circ$

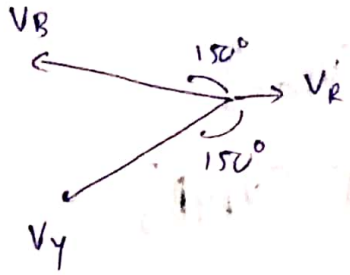
1.1)

before



$|V_R| = |V_Y| = |V_B|$

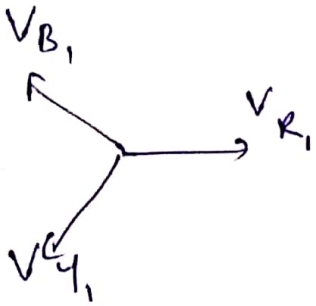
after



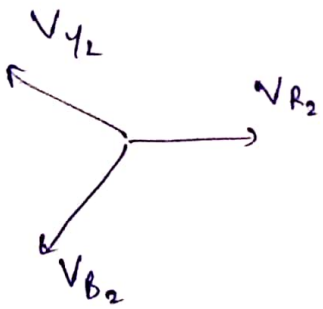
$|V_R| = |V_Y| = |V_B|$

1.2) sequence is RYB

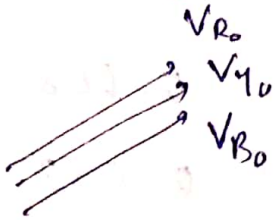
(+)



(−)



(0)



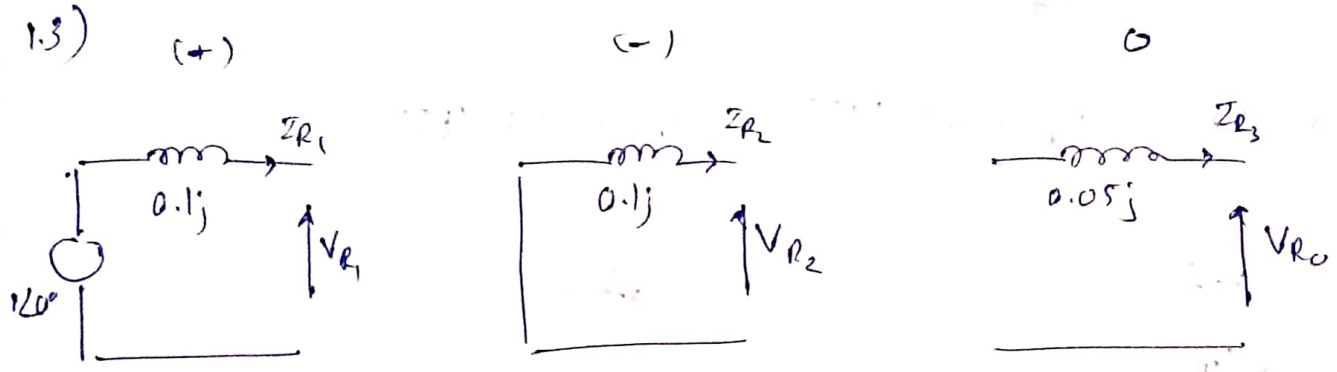
$V_R = V_{R_0} + V_{R_1} + V_{R_2}$; $\lambda = 1\angle 120^\circ$

$V_B = V_{B_0} + V_{B_1} + V_{B_2} = V_{R_0} + \lambda V_{R_1} + \lambda^2 V_{R_2}$

$V_Y = V_{Y_0} + V_{Y_1} + V_{Y_2} = V_{R_0} + \lambda^2 V_{R_1} + \lambda V_{R_2}$

$$\begin{bmatrix} V_R \\ V_Y \\ V_B \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & \lambda^2 & \lambda \\ 1 & \lambda & \lambda^2 \end{bmatrix} \begin{bmatrix} V_{R_0} \\ V_{R_1} \\ V_{R_2} \end{bmatrix} \quad ; \quad A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & \lambda^2 & \lambda \\ 1 & \lambda & \lambda^2 \end{bmatrix}$$

1.3)



$$[V_{RYB}] = A\,[V_{R012}]$$

$$[V_{R012}] = A^{-1}\,[V_{RYB}]$$

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & \lambda^2 & \lambda \\ 1 & \lambda & \lambda^2 \end{bmatrix} \quad : \quad A^{-1} = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & \lambda & \lambda^2 \\ 1 & \lambda^2 & \lambda \end{bmatrix}$$

$$\begin{bmatrix} V_{R_0} \\ V_{R_1} \\ V_{R_2} \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & \lambda & \lambda^2 \\ 1 & \lambda^2 & \lambda \end{bmatrix}\begin{bmatrix} 0.1\angle 0° \\ 1.1\angle -150° \\ 1.1\angle 150° \end{bmatrix}$$

$V_{R_0} = -0.602\angle 0°$

$V_{R_1} = 0.668\angle 0°$

$V_{R_2} = 0.033$

$1\angle 0° = 0.1j\, I_{R_1} + V_{R_1}$

$$I_{R_1} = \frac{1 - V_{R_1}}{0.1j} = -16.02j$$

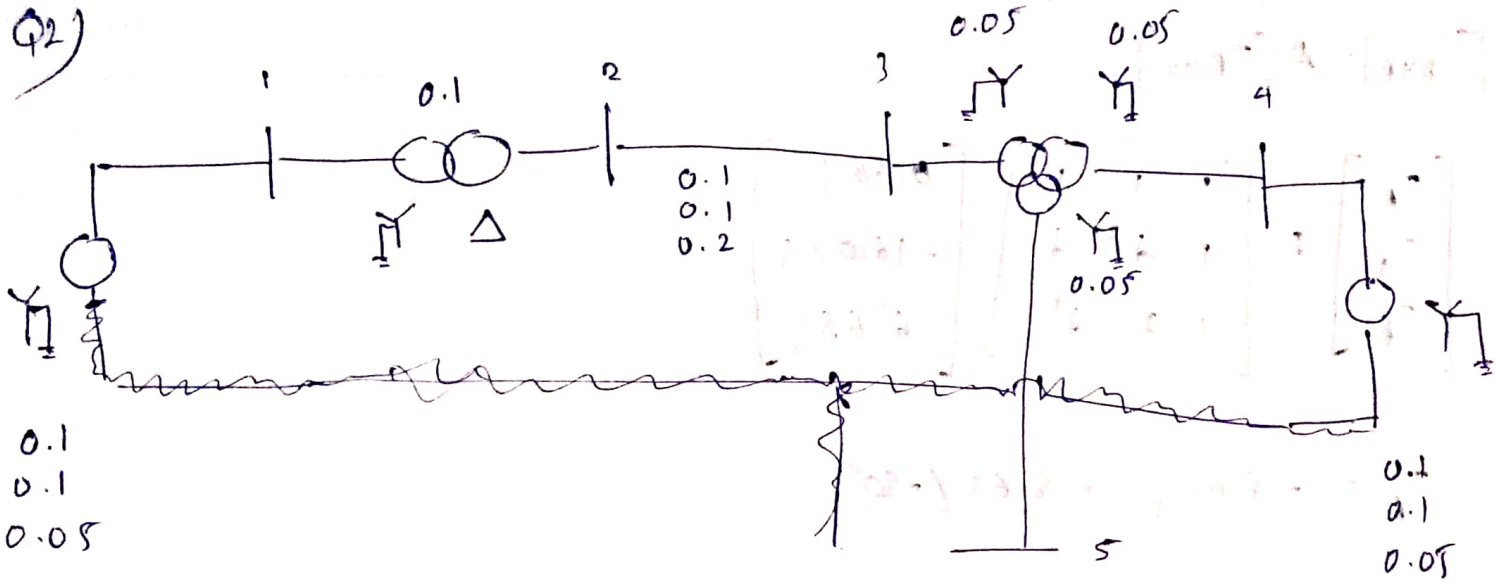
$V_{R_2} = -0.1j\, I_{R_2}$

$I_{R_2} = 6.68j$

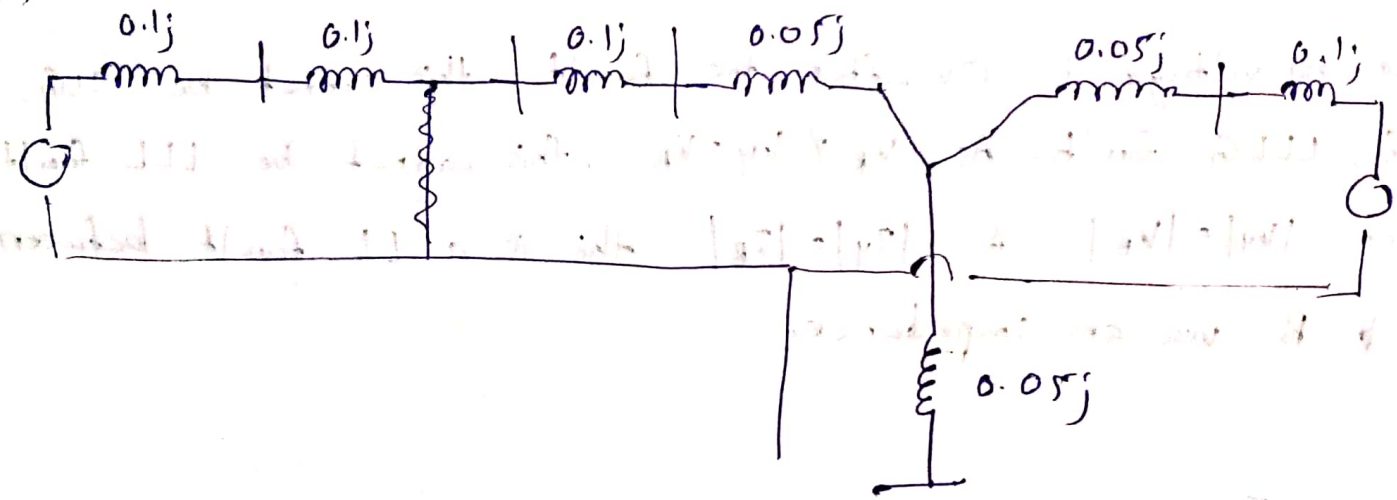
assuming generator is grounded (solidly)

$$I_{R_0} = \frac{-V_{R_0}}{0.05j} = 0.66j$$

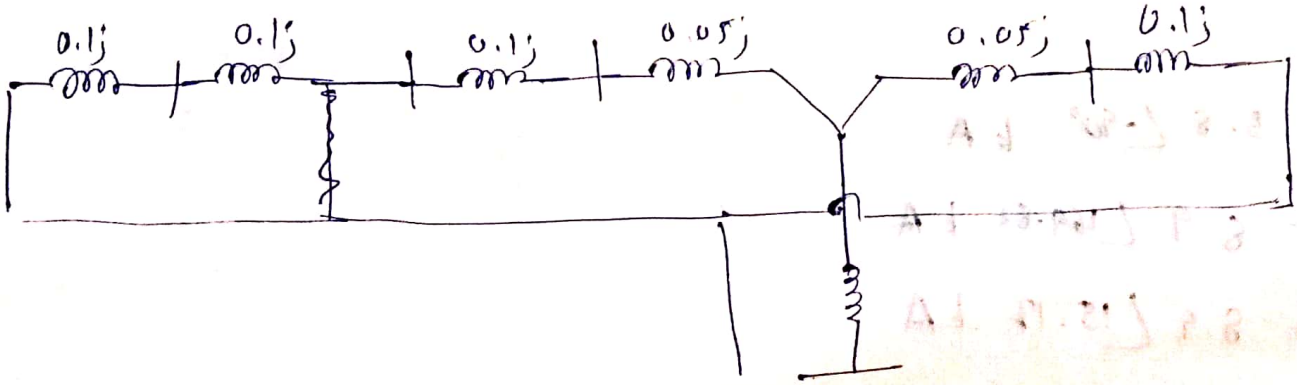
Q2)


1
2
3
4
5
0.1
0.1
0.1
0.2
0.05
0.05
0.05
0.1
0.1
0.05
0.1
0.1
0.05

(+)


0.1j
0.1j
0.1j
0.05j
0.05j
0.1j
0.05j

(−)


0.1j
0.1j
0.1j
0.05j
0.05j
0.1j

$[I_{RYB}] = A[I_{R012}]$

$$\begin{bmatrix} I_R \\ I_Y \\ I_B \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & \lambda^2 & \lambda \\ 1 & \lambda & \lambda^2 \end{bmatrix} \begin{bmatrix} 0.66j \\ -16.02j \\ 6.68j \end{bmatrix}$$

$I_R = -8.68j = 8.68 \angle -90^\circ$

$I_Y = 20.37 \angle 164.83$

$I_B = 20.37 \angle 15.17$

since no voltage is 0v after the fault, this cannot be SLG, LLG, LLLG faults. As $V_R \neq V_Y = V_B$, this cannot be LLL fault. Since $|V_Y| = |V_B|$ & $|I_Y| = |I_B|$ this is a LL fault between Y & B via an impedance

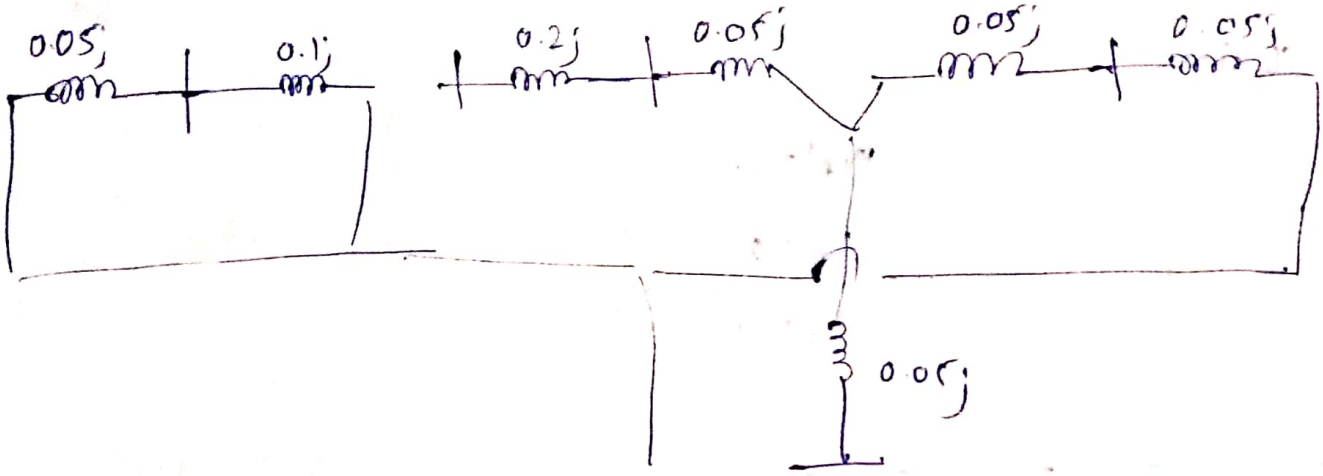
$S_{base} = \sqrt{3}\, V_{base}\, I_{base}$

$$I_{base} = \frac{100 \text{ MVA}}{\sqrt{3} \times 132 \text{ kVA}} = 0.437 \text{ kA}$$

$I_R = 3.8 \angle -90^\circ$ kA

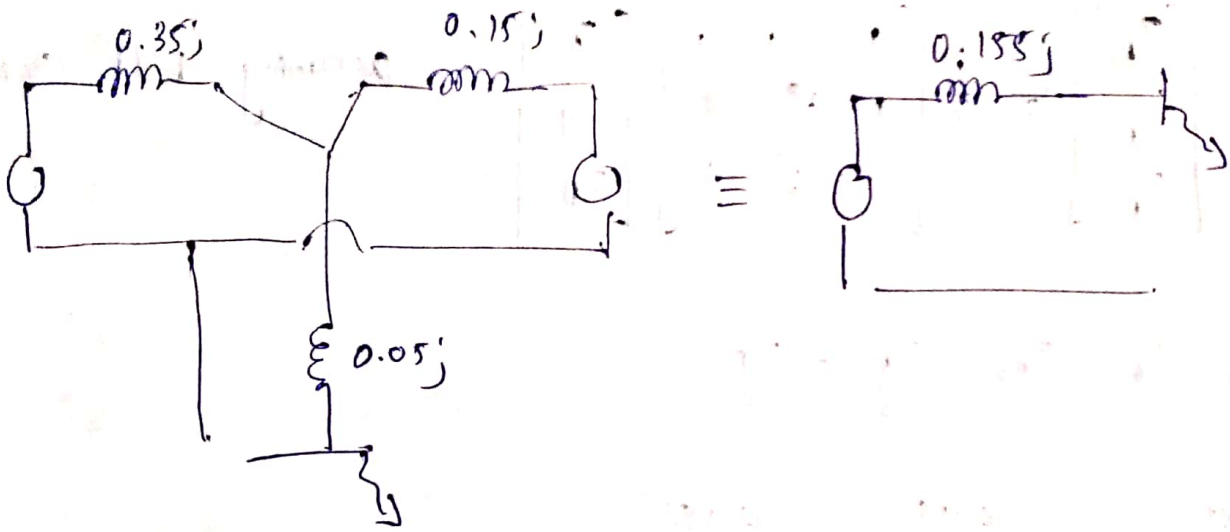
$I_Y = 8.9 \angle 164.83$ kA

$I_B = 8.9 \angle 15.17$ kA

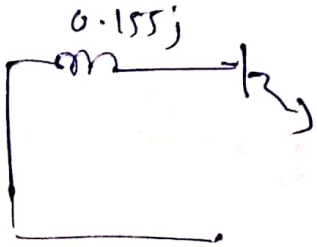

0.05j
0.1j
0.2j
0.05j
0.05j
0.05j
0.05j

2.3)

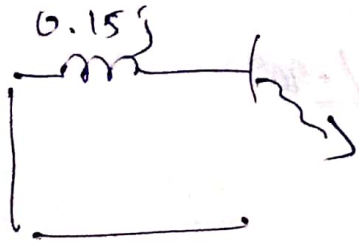
(a)


0.35j
0.15j
0.05j
≡
0.155j

(b)


0.155j

(c)


0.15j

2.4) (a) SLG

$|V_R| = 0$

$I_B = 0$

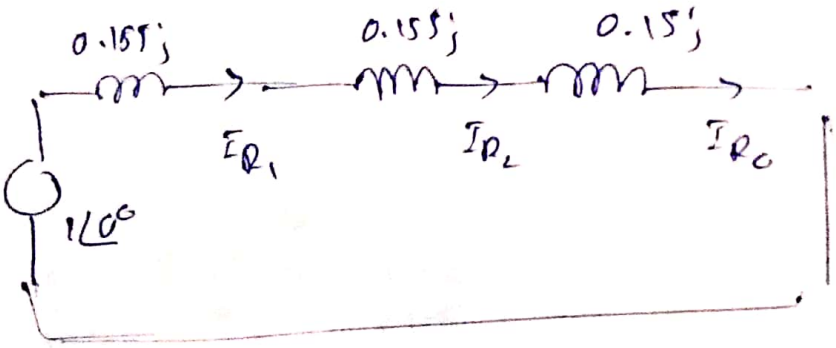
$I_Y = 0$

from $[I_{RYB}] = A\,[I_{R012}]$

$$[I_{R012}] = A^{-1}[I_{RYB}]$$

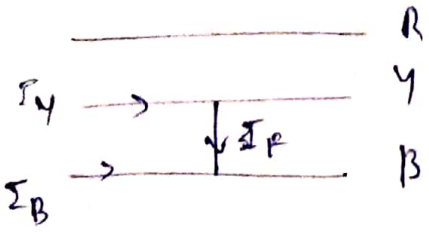
$$\begin{bmatrix} I_{R0} \\ I_{R1} \\ I_{R2} \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & \lambda & \lambda^2 \\ 1 & \lambda^2 & \lambda \end{bmatrix}\begin{bmatrix} I_R \\ I_Y \\ I_B \end{bmatrix}$$ assuming RYB sequence

$$I_{R0} = I_{R1} = I_{R2} = \frac{I_R}{3} = \frac{I_F}{3}$$



$$I_{R1} = \frac{1\angle 0^\circ}{0.15j \times 2 + 0.15j} = 2.17\angle -90^\circ$$

(b) ~~Bdo~~ LL



$|V_Y| = |V_B|$

$I_R = 0$

$I_Y = I_F = -I_B$

$$[I_{RYB}] = A[I_{R012}]$$

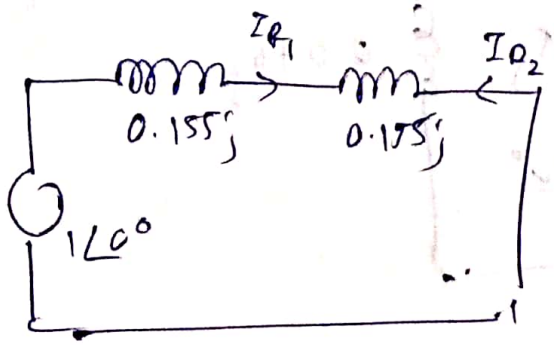
$$[I_{R012}] = A^{-1}[I_{RYB}]$$

$$\begin{bmatrix} I_{R_0} \\ I_{R_1} \\ I_{R_2} \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & \lambda & \lambda^2 \\ 1 & \lambda^2 & \lambda \end{bmatrix} \begin{bmatrix} 0 \\ I_F \\ -I_F \end{bmatrix}$$ assuming RYB sequence

$$I_{R0} = 0$$

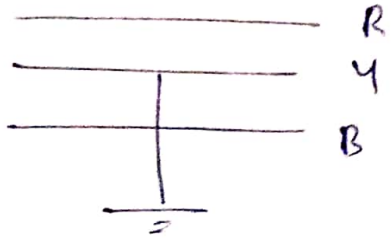
$$I_{R_1} = \frac{1}{3}(\lambda - \lambda^2) I_F$$

$$I_{R_2} = \frac{1}{3}(\lambda^2 - \lambda) I_F = -I_{R_1}$$



$$I_{R_1} = \frac{1\angle 0^\circ}{0.155j \times 2} = 3.23\angle -90^\circ$$

(c)



$|V_Y| = |V_B| = 0$

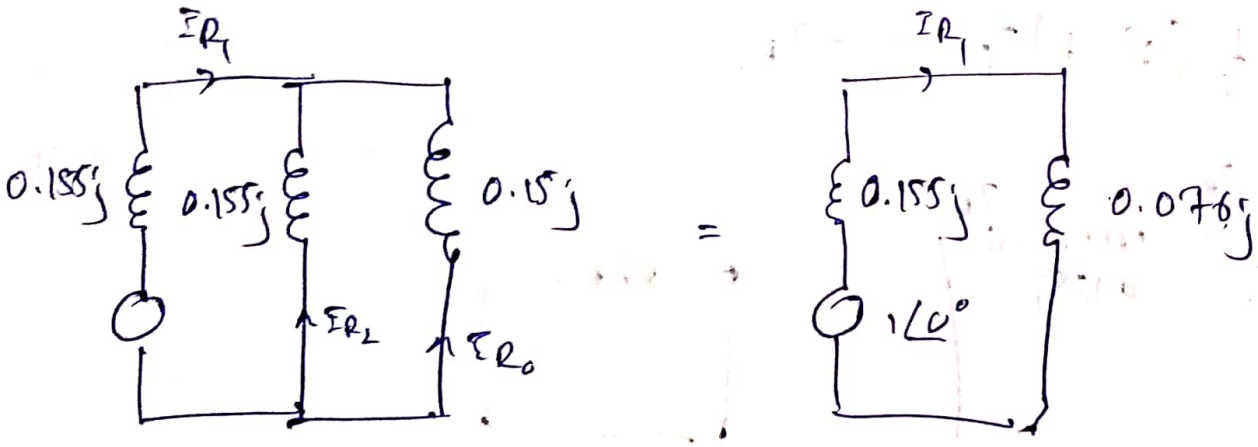
$I_R = 0$

$$[V_{RYB}] = A[V_{R012}]$$

$$[V_{R012}] = A^{-1}[V_{RYB}]$$

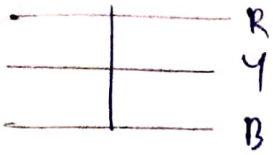
$$\begin{bmatrix} V_{R_0} \\ V_{R_1} \\ V_{R_2} \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & \lambda & \lambda^2 \\ 1 & \lambda^2 & \lambda \end{bmatrix}\begin{bmatrix} V_R \\ V_Y \\ V_B \end{bmatrix}$$ assuming RYB sequence

$$V_{R_0} = V_{R_1} = V_{R_2} = \frac{V_R}{3}$$



$$I_{R_1} = \frac{1\angle 0°}{0.155j + 0.076j} = 4.33\angle -90°$$

(d)



$|V_R| = |V_Y| = |V_B|$

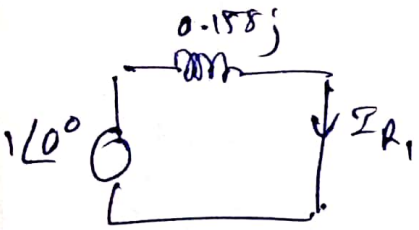
$[V_{RYB}] = A[V_{R012}]$

$[V_{R012}] = A^{-1}[V_{RYB}]$

$$\begin{bmatrix} V_{R0} \\ V_{R1} \\ V_{R2} \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & \lambda & \lambda^2 \\ 1 & \lambda^2 & \lambda \end{bmatrix}\begin{bmatrix} V_R \\ V_Y \\ V_B \end{bmatrix}$$

$V_{R_1} = 0 = V_{R_2}$

$V_{R0} = V_R$

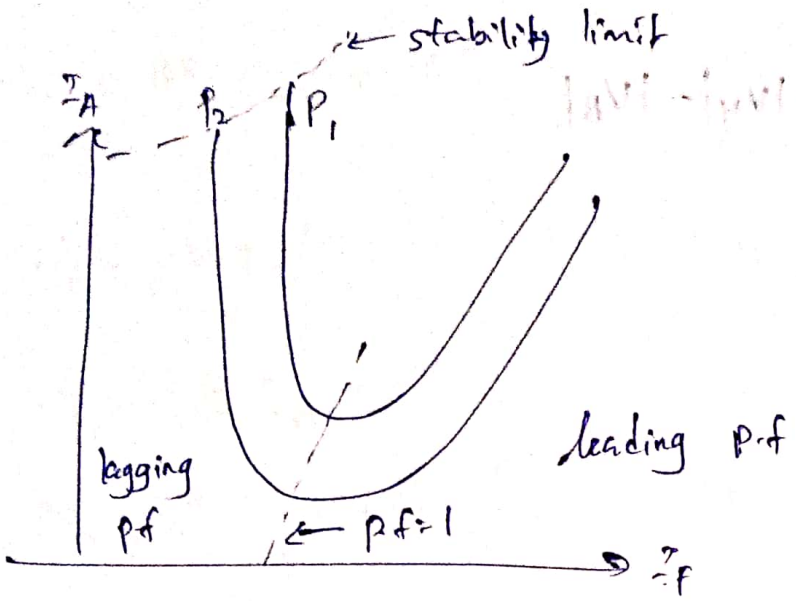


$$I_{R_1} = \frac{1\angle 0^\circ}{0.155j} = 6.45\angle -90^\circ$$

Q3)

3.1)

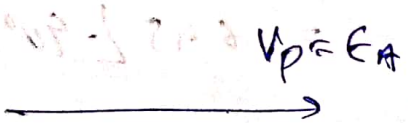
(a)



(b) A synchronous motor without the rotor shaft is a synchronous condenser.

This is used to supply or consume reactive power.

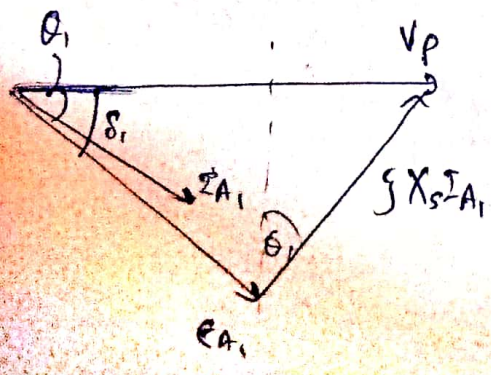
(c) Since synchronous condenser is synchronous motor at no load, no active power.



3.2) Δ, $V_p = 230V$, $S = 50\ kVA$, $P_o = 32\ hp = 23.872\ kW$

$\approx 23.9\ kW$

$X_s = 2.75\ \Omega$, $P_{loss} = 3.5\ kW$



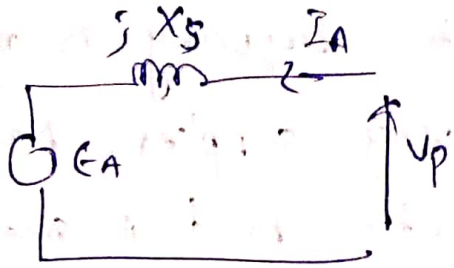
$P_{in} = P_o + P_{loss} = 27.4\ kW$

$P_{in} = \sqrt{3}\, V_L I_L \cos\theta_1$

$I_L = \dfrac{27.4\ kW}{\sqrt{3} \times 230 \times 0.82} = 83.79$

$I_L \approx 83.8 \angle -34.92^\circ$

$I_A = \frac{I_L}{\sqrt{3}} = 48.4 \angle -34.92^\circ$



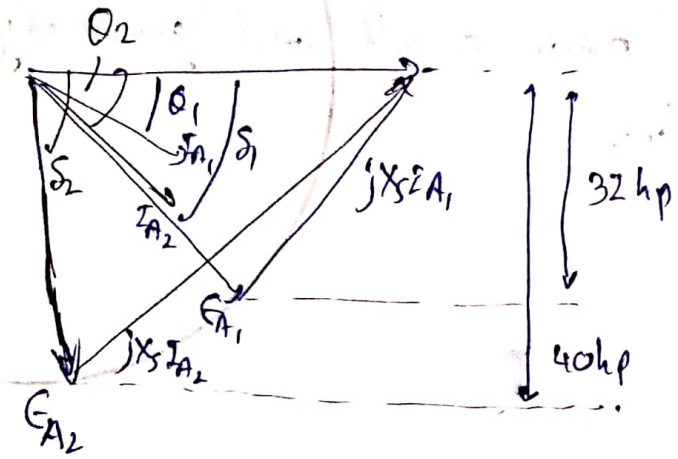
$V_p = E_A + jX_s I_A$

$E_A = V_p - jX_s I_A$

$= 230 \angle 0^\circ - j\,2.75 \times 48.4 \angle -34.92^\circ$

$= 188.6 \angle -35.36^\circ$

(b)



$\frac{E_{A_1} \sin\delta_1}{E_{A_2} \sin\delta_2} = \frac{32}{40} \longrightarrow \sin\delta_2 = \frac{40}{32}\sin\delta_1 = 0.723$

$\delta_2 = 46.34^\circ$

$E_{A_2} = 188.6 \angle -46.34^\circ$

$V_p = E_{A_2} + jX_s I_{A_2}$

$I_{A_2} = \frac{V_p - E_{A_2}}{jX_s} = \frac{230 - 188.6 \angle -46.34^\circ}{j \times 2.75} = 61.47 \angle -36.18^\circ$

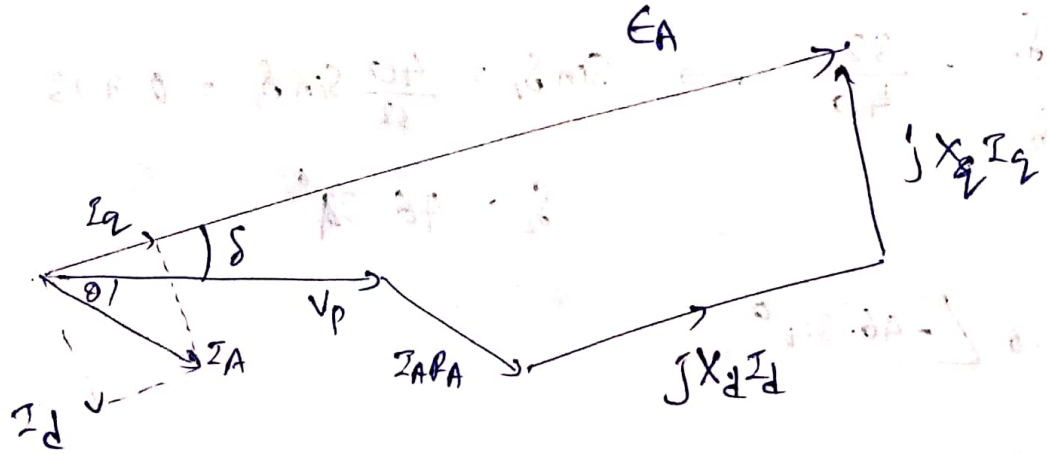
3.3) Stator frequency is reduced initially using a variable frequency supplier. Now rotational speed of stator magnetic field is small, therefore rotor magnetic field can follow it. Gradually increase the stator frequency to rated frequency.

As the motor speed is low, internally generated voltage ~~of~~ is small. With this, from $V_p = E_A + jX_S I_A$, I_A will be large if connected to ~~an~~ utility. ~~So~~ Therefore V_p must be made small and gradually increased to rated value.

Q4

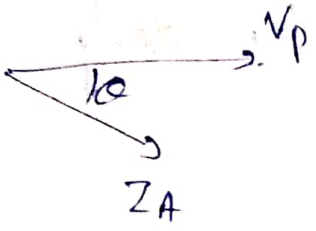
4.1) Consider lagging p.f.

(a)

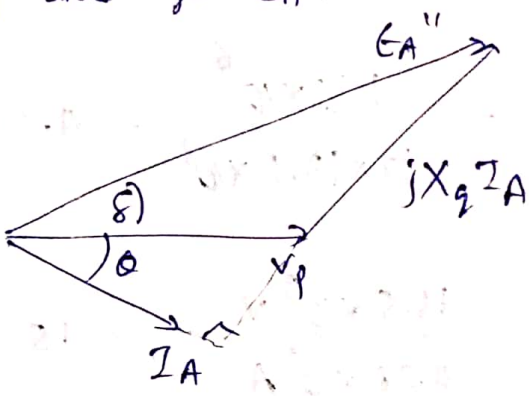


(b) we know V_p, I_A, θ. Assuming X_q, X_d are given.

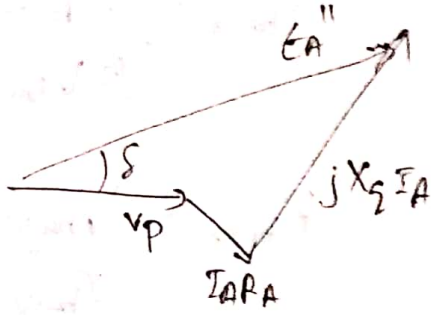
step 1: draw V_p and I_A



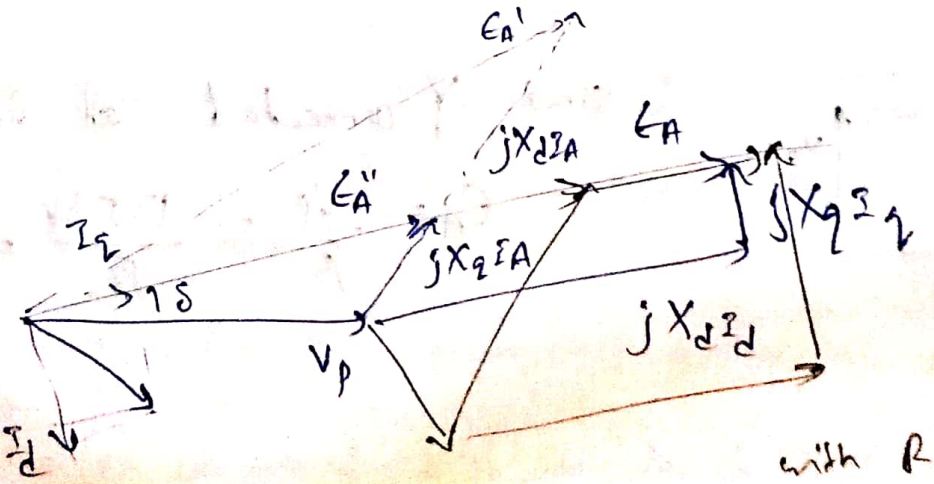
step 2: Neglecting R_A ($\ll X_q, X_d$), assuming this as a cylindrical rotor ~~motor~~ with $X_s = X_q$, draw $jX_q I_A$ and get E_A''.



with R_A



step 3: Now we can find δ. Then we can find I_q & I_d. After finding I_q & I_d, draw $jX_q I_q$ & $jX_d I_d$ to find E_A



with R_A

4.2)

(a) ~~ac compol~~ • unidirectional component which decays exponentially in few cycles.

• fundamental frequency component which decays very rapidly initially and then slowly to a steady state value.

(b) $$I(t) = (I'' - I')e^{-\frac{t}{T''}} + (I' - I_{ss})e^{-\frac{t}{T'}} + I_{ss}$$

(c) $S_{base} = \sqrt{3}\, V_{base} I_{base}$

$$I_{base} = \frac{S_{base}}{\sqrt{3}\, V_{base}} = \frac{100 \times 10^6}{\sqrt{3} \times 13.5 \times 10^3} = 4.28 \text{ kA}$$

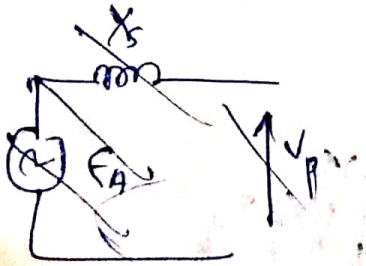
$$Z_{base} = \frac{V_{base}}{I_{base}} = \frac{13.5 \times 10^3 \text{ V}}{4.28 \times 10^3 \text{ A}} = 3.15\,\Omega$$

actual values

$X_s = 3.15 \times 1 = 3.15\,\Omega$

$X' = 0.3 \times 3.15 = 0.95\,\Omega$

$X'' = 0.15 \times 3.15 = 0.47\,\Omega$

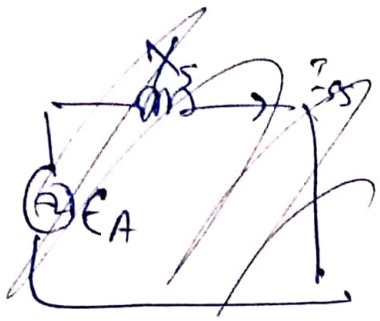


since Y connected at no load

$$E_A = \frac{V_p}{\sqrt{3}} = \frac{13.5 \text{ kV}}{\sqrt{3}} = 7.79 \text{ kV}$$

$$X'' = \frac{E_A}{I''} \longrightarrow I'' = \frac{7.79\ kV}{0.47\ \Omega} = 16.58\ kA$$

$$X' = \frac{E_A}{I'} \longrightarrow I' = \frac{7.79\ kV}{0.95\ \Omega} = 8.2\ kA$$



$$X_s = \frac{E_A}{I_{ss}} \longrightarrow I_{ss} = \frac{7.79\ kV}{3.15\ \Omega} = 2.47\ kA$$

$$I(t) = (16.58 - 8.2)\, e^{-\frac{t}{T''}} + (8.2 - 2.47)\, e^{-\frac{t}{T'}} + 2.47$$

$$I(t) = \left(8.38\, e^{-\frac{t}{0.03}} + 5.73\, e^{-\frac{t}{1.15}} + 2.47\right)\ kA$$

4.3

Advantages - Small rotor speed variations (in 2 %)
typical o/p is 690 V_{LL} V

Disadvantages - As ω constant, can't capture maximum power for different wind speeds as optimum C_p for different wind speeds can't be realized.

Variants - Operate generator in 2 wind speeds by connecting generators with different number of poles to the same gear box.

- Use wound rotor type induction generator to have variable slip operation.
speed range of about 10% can be achieved.

2017

Q3)

3.1) 4 parameters have to be checked,

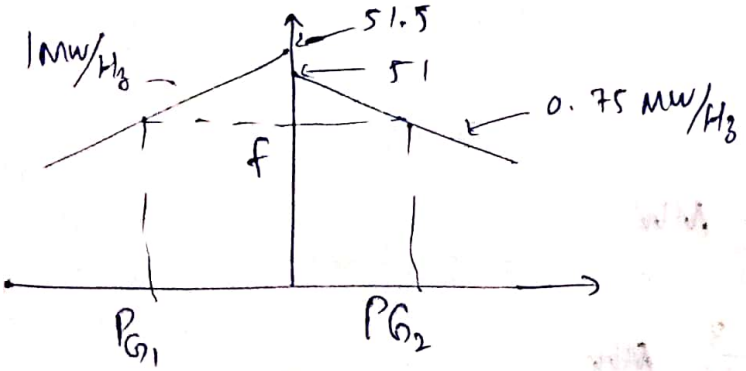
- voltage magnetude
- phase sequence
- phase difference
- frequency.

following condutions must be satisfied

Generator		power system
RMS voltage	=	RMS voltage
phase sequence	=	phase sequence
phase difference	=	phase difference
frequency	>	frequency

3.2)

a) active power = 2.5 MVA × Cos θ = 2.5 × 0.8 = 2 MW



$P_{G_1} + P_{G_2} = 2$ MW

$0.75 = \frac{P_{G_2}}{51 - f} \rightarrow f = 51 - \frac{P_{G_2}}{0.75}$

$1 = \frac{P_{G_1}}{51.5 - f} \rightarrow f = 51.5 - P_{G_1}$

$51 - \frac{P_{G_2}}{0.75} = 51.5 - P_{G_1}$

$P_{G_1} - \frac{P_{G_2}}{0.75} = 0.5$

$2 - P_{G_2} - \frac{P_{G_2}}{0.75} = 0.5 \rightarrow P_{G_2} = \frac{9}{14}$ MW

$P_{G_1} = \frac{19}{14}$ MW

$f = 50.143$ Hz

(b) $P_{G_1} + P_{G_2} = P$

$$1 = \frac{P_{G_1}}{51.5 - f} \qquad 0.75 = \frac{P_{G_2}}{51 - f}$$

$$51.5 - P_{G_1} = f \qquad f = 51 - \frac{P_{G_2}}{0.75} = 51 - \frac{(P - P_{G_1})}{0.75}$$

$$51.5 - P_{G_1} = 51 - \frac{P}{0.75} + \frac{P_{G_1}}{0.75}$$

$$0.5 + \frac{P}{0.75} = \frac{1.75\, P_{G_1}}{0.75} \rightarrow P_{G_1} = \frac{0.375 + P}{1.75}$$

$\boxed{8 \sim 9}$

$$P = 1.5 \text{ MW}$$

$$P_{G_1} = \frac{0.375 + 1.5}{1.75} = \frac{15}{14} \text{ MW}$$

$$P_{G_2} = 1.5 - P_{G_1} = \frac{3}{7} \text{ MW}$$

$$f = 50.43 \text{ Hz}$$

$\boxed{9 \sim 10}$

$$P = 1 \text{ MW}$$

$$P_{G_1} = \frac{0.375 + 1}{1.75} = \frac{11}{14} \text{ MW}$$

$$P_{G_2} = 1 - P_{G_1} = \frac{3}{14} \text{ MW}$$

$$f = 51.5 - P_{G_1} = 50.71 \text{ Hz}$$

$\boxed{10\sim 11}$

$P = 0.5\ MW$

$P_{G_1} = \dfrac{0.375 + 0.5}{1.75} = 0.5\ MW$

$P_{G_2} = P - P_{G_1} = 0\ MW.$

$f = 51.5 - P_{G_1} = 51\ Hz.$

$\boxed{11\sim 12}$

$P = 0.4\ MW$

$P_{G_1} = \dfrac{0.375 + 0.4}{1.75} = \dfrac{31}{70}\ MW$

$P_{G_2} = P - P_{G_1} = -\dfrac{3}{70}\ MW$ ← Generator 2 ~~can~~ acts as a motor. ∴ will be disconnected.

~~$f = 51.5 - \frac{31}{70} = 51.06\ Hz.$~~

all power supplied by ~~P~~ generator 1.

$\therefore P_{G_1} = 0.4\ MW.$

$f = 51.5 - 0.4 = 51.1\ Hz.$

	Generator 1	Generator 2
8~9	Provide power	Provide power
9~10	Provide power	Provide power
10~11	Provide power	Provide power
11~12	Provide power	Disconnected

(c) New $P = 3\,MW - 1.5\,MW = 1.5\,MW$

$$P_{G_1} = \frac{0.375 + P}{1.75} = \frac{0.375 + 1.5}{1.75} = \frac{15}{14}\,MW$$

$$P_{G_2} = P - P_{G_1} = 1.5 - \frac{15}{14} = \frac{3}{7}\,MW$$

$$f = 51.5 - P_{G_1} = 50.43\,Hz$$

we have to increase the no load frequency of

(a) Generator 1 such that all power is supplied by generator 1

or

(b) Generator 2 such that power is shared between 2 generators.

This can be achieved by a means of a governer.

Q4

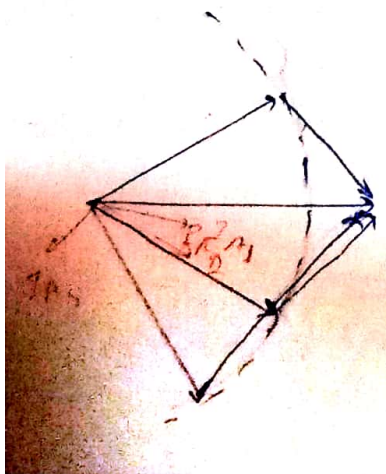
4.1) Since stiff voltage source V_p constant

~~since~~ $E_A = k\phi\omega$; If constant means ϕ constant

assuming f is constant ω constant

$\therefore |E_A|$ is constant

$X_s I_A \cos\theta$ is proportional to active power



4.2) $P_o = 16\ hp \approx 12\ kW$

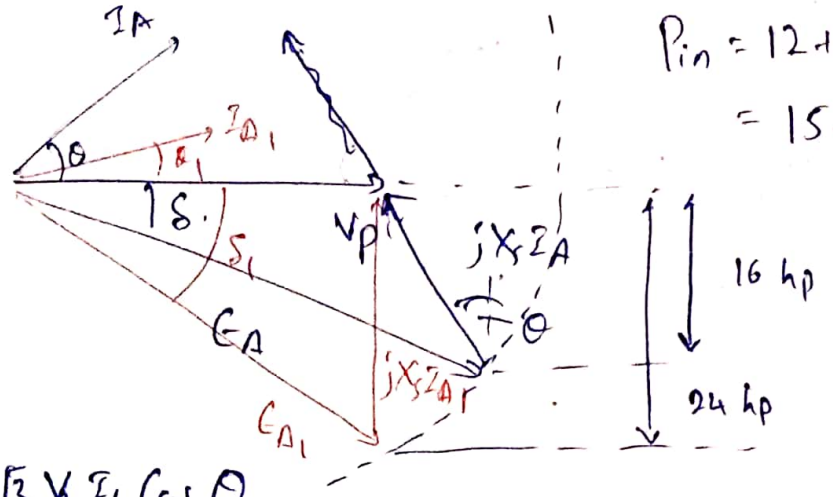
$P_{F,W} = 2\ kW$

$P_{core} = 1.25\ kW$

$P_{in} = P_o + P_{F,W} + P_{core} + P_{SCL} + P_{stray}$

$P_{in} = 12 + 2 + 1.5$

$= 15.5\ kW$



$P_{in} = \sqrt{3}\, V_L I_L \cos\theta$

$15.5\ kW = \sqrt{3} \times 230 \times I_L \times 0.85$

$I_L = 45.77 \angle 31.79^\circ\ A$

$I_A = \dfrac{I_L}{\sqrt{3}} = 26.43 \angle 31.79^\circ\ A$

$V_p = E_A + jX_sI_A$

$E_A = V_p - jX_sI_A = 230 - j \times 2.75 \times 26.43 \angle 31.79^\circ$

$= 275.31 \angle -12.97^\circ\ V$

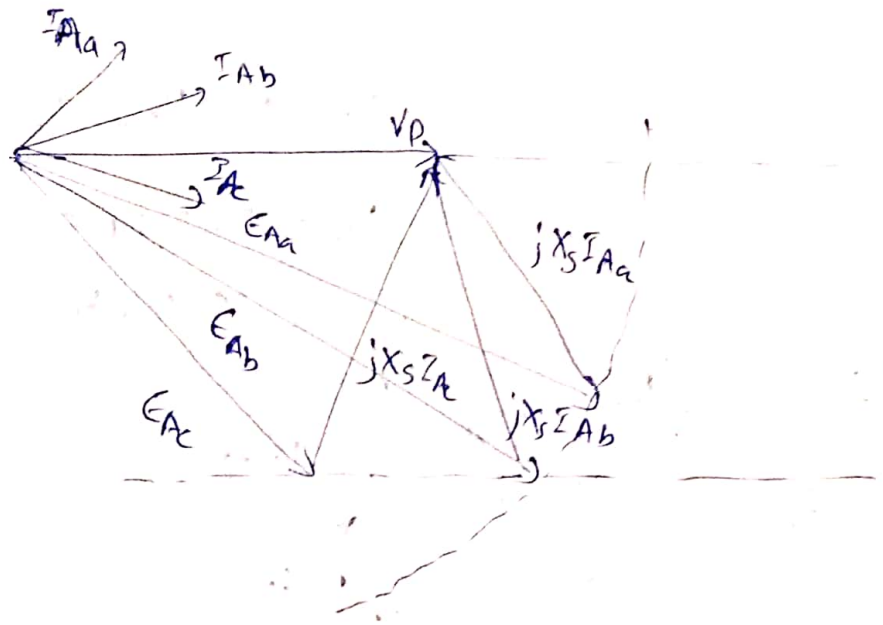
(b) $\dfrac{E_A \sin\delta}{E_{A_1} \sin\delta_1} = \dfrac{16}{24} \rightarrow \sin\delta_1 = \dfrac{24}{16}\sin\delta = 0.337 \rightarrow \delta_1 = 19.67^\circ$

$E_{A_1} = 275.31 \angle -19.67^\circ\ V$

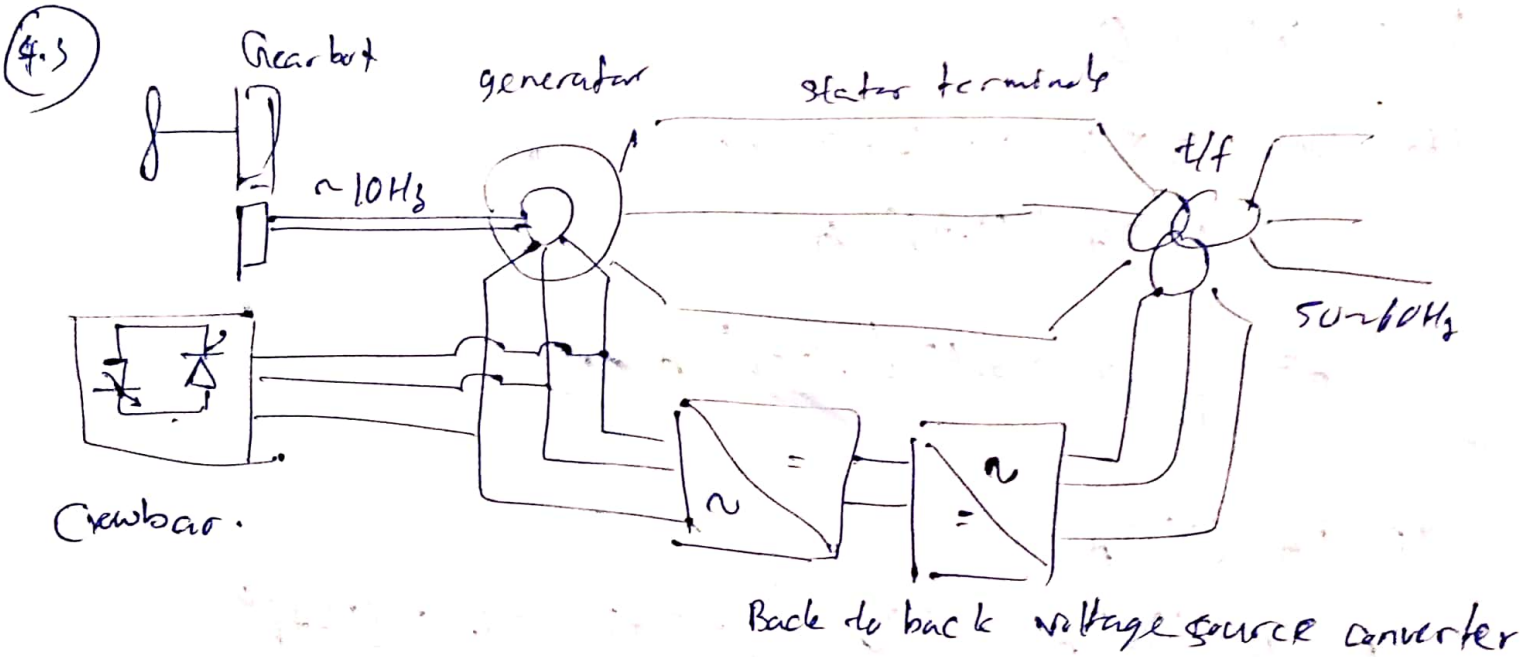
$V_p = E_{A_1} + jX_sI_{A_1} \rightarrow I_{A_1} = \dfrac{V_p - E_{A_1}}{jX_s} = 35.34 \angle 17.51^\circ\ A$

$\therefore I_L = \sqrt{3}\, I_A = 61.2 \angle 17.51^\circ$ A

(c)



(4.3)



Back to back VSC are used to match frequency and voltage of grid.

Crowbar is used for protection.

Generator is wound rotor type as access to rotor is required.

Can ~~be~~ transfer power through stator and rotor.